

Pedaling into Presence: Evaluating Presence and User Experience in VR Cycling vs. Controller Use

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ABSTRACT

This study examined two input methods - real bicycle and gaming controller - in a virtual bike tour. Participants engaged in a virtual bike tour using both methods while assessing presence and user experience. The bicycle method notably enhanced sense of physical space, engagement, and ecological validity, resulting in a more immersive experience. While pragmatic qualities showed no significant differences, the bicycle method excelled in hedonic qualities. The findings indicate that realistic physical interaction boosts presence and appeal, suggesting future research on different movement types.

Index Terms: Human-centered computing → Human computer interaction (HCI) → Interaction devices → Haptic devices

1 INTRODUCTION & STATE OF THE ART

There is a variety of locomotion techniques to move in virtual worlds, which can be classified according to Boletsis (2017) [1] as physical and artificial. Implementing a consistent locomotion method for VR and reality is challenging due to technical, spatial, and sensory issues, as virtual worlds can exceed real-world space. While movement between reality and VR enhances presence and reduces motion sickness, artificial locomotion methods such as teleportation also achieve good results in these areas [2]. However, when looking at factors such as spatial orientation or sensation of self-motion, it quickly becomes apparent that real movement has a positive effect on these [2]. An alternative to real walking is pedaling or riding a bicycle, as this also involves real movement, but takes up less space than real walking. Studies show a positive influence of pedaling movements on the perceived sense of presence, distance estimation and feeling of control [3] as well as on ownership, agency, and walking sensation [4]. Another study by Chen et al. (2024) [5] also investigated the influence of real pedaling vs. controller, but with a senior sample and proved the positive influence of cycling on spatial orientation, safety and enjoyment. In these studies, however, the pedal movement was transferred to a walking movement in the virtual environment or the sample was very specific, which means that no general statements can be made. Research is still necessary to determine if real cycling effects can be replicated in VR for a general audience.

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2 EXPERIMENT

A study was designed to explore different input methods for a virtual bicycle tour. Control was facilitated using a real bicycle and a traditional gaming controller (see Figure 1). In both conditions, the participants completed a virtual bicycle tour along a lake, which included uphill and downhill sections as well as straight sections.



Figure 1: View in the VR scene (left) and setup for the two conditions (right)

In order to integrate the real pedaling and steering movements of the bicycle into the virtual scenario, the bicycle was combined with a roller trainer and a rotating front wheel support. When the user pedals, the movement on the roller trainer is sent via radio transmission (ANT or ANT+ protocol) to the corresponding virtual scenario, in which the translation speed is calculated. Virtual uphill and downhill rides along the route were simulated by adaptive application of positive or negative acceleration to the rear tire of the bike, depending on the incline of the simulated terrain by the bicycle home trainers' computer-controlled force feedback motor. This made it possible to realistically simulate gradients of up to 20 percent and descents of up to 5 percent. In a within-subject study, the two conditions (controller vs. bicycle) were compared in terms of presence and user experience. The perceived presence was assessed with a short version of the ITC-SOPI, which included only 12 instead of 44 items to measure the factors sense of physical space, engagement, ecological validity and negative effects [6]. User Experience was measured by the User Experience Questionnaire (UEQ) [7], which contains 26 bipolar items divided into 6 scales: attractiveness, perspicuity, efficiency, dependability, stimulation and novelty. The conditions were tested while seated (real bike vs. chair) and with a HTC-Vive and in counterbalanced order. 20 people (11 women, 9 men) with an average age of 23.85 (SD= 4.63) took part in the study. All had normal or corrected-to-normal vision, participated in the study voluntarily and were allowed to abort the experiment at any time.

Table 1: Results of the Wilcoxon-test for both conditions

	Sense of physical space	Engagement	Ecological validity	Negative effects	Attractiveness	Perspicuity	Efficiency	Dependability	Stimulation	Novelty
Mdn bicycle	2.33	2.66	2.33	1.33	0.75	2.12	0.75	1.12	1.25	0.75
Mdn controller	1.83	1.66	1.83	1.83	0.08	2.75	0.75	1.25	0.62	0.50
z	-2.05	-2.39	-2.16	1.46	1.97	-1.36	0.86	-0.86	1.67	1.11
p	.040	.017	.031	.145	.049	.175	.390	.390	.095	.265
r	0.46	0.53	0.48	0.29	0.43	0.30	0.19	0.19	0.37	0.24

3 RESULTS

In order to answer the question of whether the type of interaction medium has an influence on presence and user experience, the two conditions - bicycle vs. controller - were compared with regard to the results for the factors.

The mean values and standard deviations for the two conditions of the presence factors are shown in Figure 2 and for the user experience in Figure 3. After a Shapiro-Wilk test showed a violation of the normal distribution, a Wilcoxon-test was conducted and showed, that the bicycle condition had significantly higher values for sense of physical space, engagement, ecological validity and attractiveness with medium to high effect sizes (according to [8]) (see table 1 for detailed results).

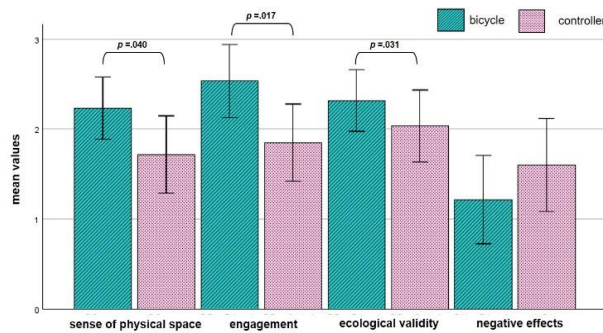


Figure 2: Means of the presence factors for both conditions

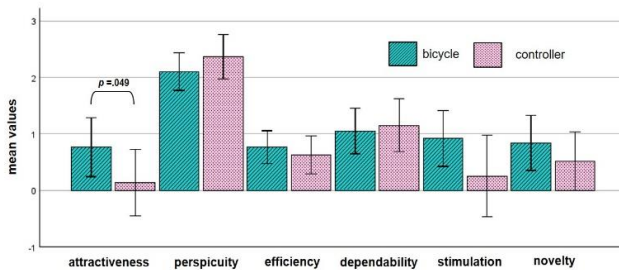


Figure 3: Means of the UEQ factors for both conditions

4 DISCUSSION AND CONCLUSION

The results show a clear increase in the presence factors sense of physical space, engagement and ecological validity for the bicycle condition. The real and consistent interactions in both virtual and real environments heightened users' presence in the virtual space, enhancing the scenario's naturalness and believability, leading to greater user interest and involvement. This increased engagement is also evident in the attractiveness factor. However, in the

investigated scenario the kind of interaction has no significant effect on the pragmatic (perspicuity, efficiency, dependability) and hedonic (stimulation, novelty) quality, even though the results showed higher values for hedonic quality in the bicycle condition. For negative effects, there is no significant result, but the bicycle condition achieved lower values for this factor. Thus, the real movement contributes positively to the reduction of this effect. This is important in situations where quick transportation with minimal negative impacts is needed. Consistent interaction positively affects presence and scenario attractiveness. Future research should explore this effect across different movement types.

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